

Less theater, more engineering

What 80% AI accuracy actually costs.



THE PREMISE

The 80% number hides the engineering problem

A generated artifact can be right and the delivery system can still be unsafe. The questions that decide whether you ship are operational, not statistical.

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- 01 What did the agent see?
 - 02 What tools did it call?
 - 03 What changed?
 - 04 Who approved the risk?
 - 05 Can we replay or roll back the decision?
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ARTIFACT GENERATION $\neq$ TRUSTWORTHY DELIVERY

THE SYSTEM

An agentic SDLC is a chain of decisions

01



Specify

requirements
+ constraints

02



Plan

tickets
+ acceptance criteria

03



Build

code + config

04



Verify

tests + evidence

05



Operate

release
+ rollback

The unit of governance is **the chain**, not the code file.

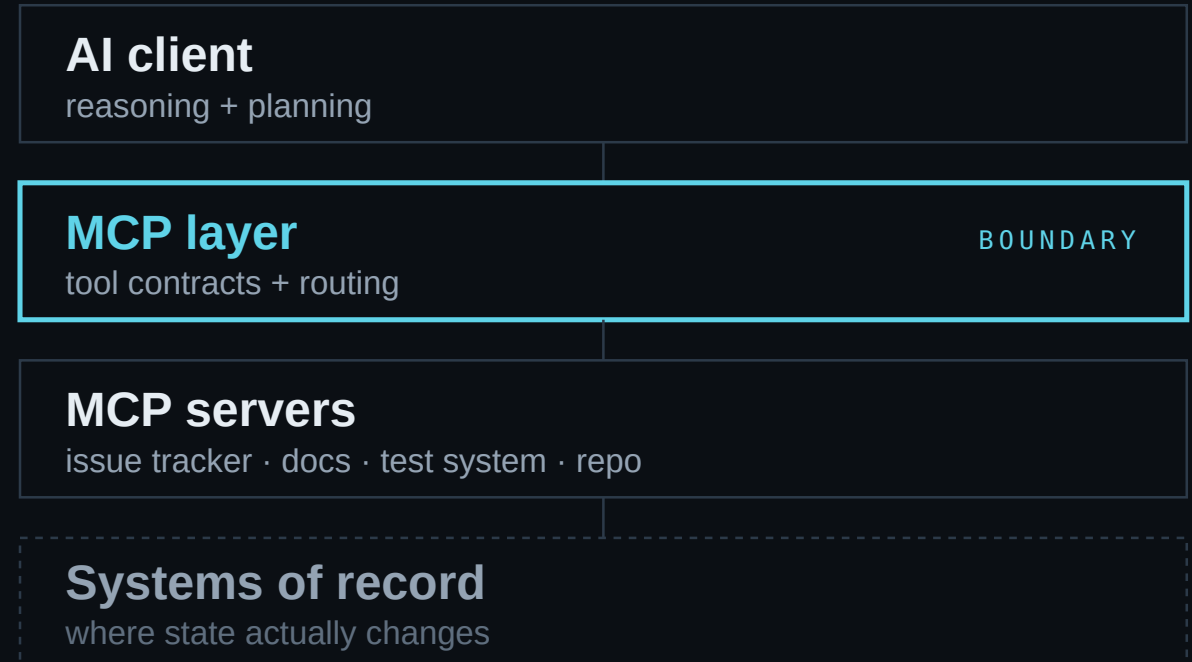
THE ARCHITECTURE

MCP makes the pipeline composable

A standard layer between an AI client and the tools that change state. Every tool contract is also a governance boundary — the place where you decide what the agent may touch.

Composability increases leverage. It also increases **blast radius**.

LAYER



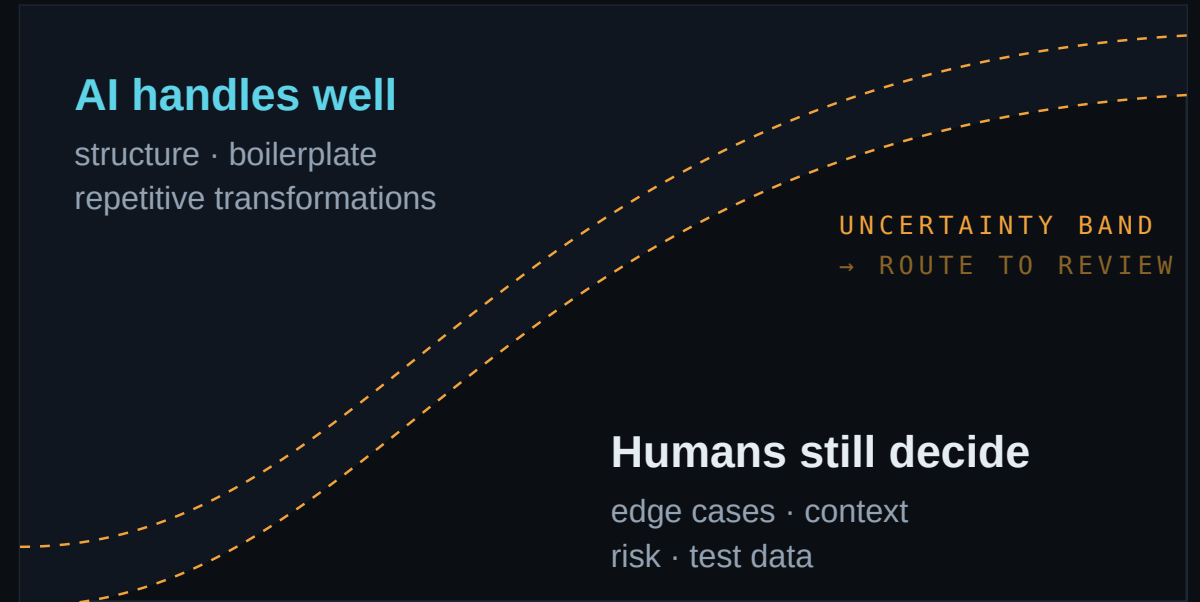
The useful output is an evidence trail

For every state-changing action, preserve enough context to reconstruct the decision months later — by someone who was not in the room.

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- 01 What did the agent receive?
 - 02 Which model and tool version ran?
 - 03 What permission was used?
 - 04 What evidence supported the result?
 - 05 What did a human accept, reject, or roll back?
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The last 20% is where quality engineering lives

The goal is not removing judgment. It is moving judgment to the cases where **uncertainty is highest**.



WHAT BREAKS

Failure modes are predictable



Sparse specification

vague acceptance criteria, nothing to evaluate against

Hallucinated interfaces

plausible names, wrong APIs

Locator or schema drift

tests keep passing and silently lose their meaning

Context loss

later steps cannot reconstruct earlier decisions

Most failures are specification, state, or evidence failures **before** they are model failures.

Risk tiers decide where humans sit

Low risk does not mean zero risk. It means the consequence is bounded and the permissions are narrow. Reversibility and business impact set the tier.

HIGH

production config · payments · auth — approval + least privilege

MEDIUM

dependency updates · regression — review + rollback

LOW

documentation · unit tests — automated checks

Measure rework, not generation

Unsafe tool-call rate

Did the agent attempt an action outside policy?

Rollback frequency

How often did a human or system reverse the change?

Defect escape rate

What reached the next stage, or production?

Audit-trail completeness

Can the action be reconstructed later?

Four operational measures turn governance into **analytics** — and analytics into an autonomy threshold you can move on evidence.

WHEN TO USE IT

A simple go / no-go test

USE THE PIPELINE WHEN

- The specification is rich enough to evaluate against
- A reviewer has time to inspect uncertainty
- The target stack is represented in the tools
- The change can be rolled back

DO NOT, WHEN

- The specification is still being discovered
- The path is safety-critical or highly regulated
- Legacy integration is poorly understood
- No senior reviewer owns the final decision

Sometimes the right engineering answer is **not yet**.

Three ideas to take back to your team

The best agentic pipeline is not the one that generates the most. It is the one that makes **uncertainty visible**.

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- 01 Treat the chain as the system under test
 - 02 Preserve evidence at every state-changing step
 - 03 Give autonomy only where the risk data supports it
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